COLLEGE OF INFORMATION SCIENCE

CYBR312
Python Scripting in Cybersecurity
Week 5: Building a Mini Rainbow Table

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Week Five Agenda

Week 5

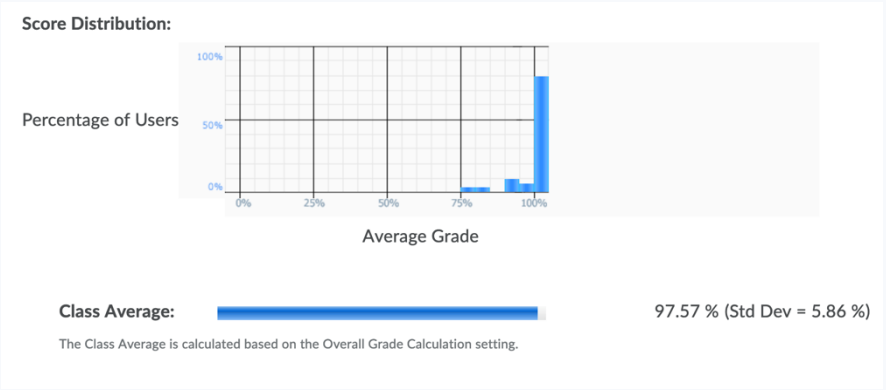
- Quiz Review
- WK-4 Solution
- Creating a mini-rainbow table using a dictionary
- Serialize the rainbow dictionary
- Create a mini-rainbow table for your assignment



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Week 4 Quiz Results

Use this slide as a visual reference during the lecture.



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WK-4 SOLUTION DEMO

Use this slide as a visual reference during the lecture.



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Week Five Assignments

Week 5

Your To Do List

- Attend: Week Five Live Lecture
- Quiz 5: Rainbow Table
- Assignment 5: Build a mini-rainbow

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Videos

Week 5

Week Five Videos

This week's video will be presented live during the Webinar.

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Key Ideas

Week 5

- Topics
- Introducing the Python itertools library
- What is a rainbow table?
- Creating a mini-rainbow dictionary
- Serialize the resulting rainbow table

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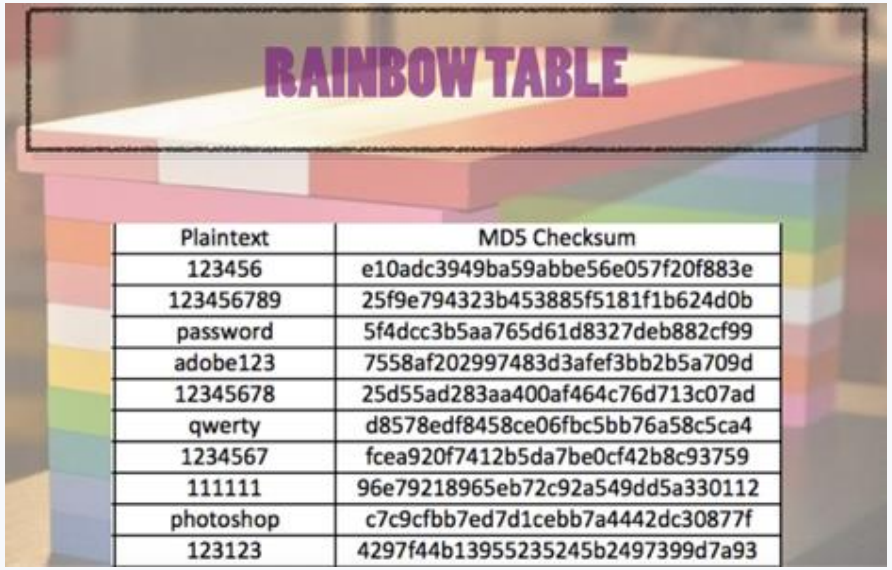
What is a rainbow table?

- Rainbow Table
- A precomputed table containing a cryptographic hash value associated with each unique password.
- They are used for cracking passwords
- In our case we will limit the set of characters and length of passwords in order to reduce the size of the table.

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What is a rainbow table?

Use this slide as a visual reference during the lecture.



Plaintext	MD5 Checksum
123456	e10adc3949ba59abbe56e057f20f883e
123456789	25f9e794323b453885f5181f1b624d0b
password	5f4dcc3b5aa765d61d8327deb882cf99
adobe123	7558af202997483d3afef3bb2b5a709d
12345678	25d55ad283aa400af464c76d713c07ad
qwerty	d8578edf8458ce06fbc5bb76a58c5ca4
1234567	fcea920f7412b5da7be0cf42b8c93759
111111	96e79218965eb72c92a549dd5a330112
photoshop	c7c9cfbb7ed7d1cebb7a4442dc30877f
123123	4297f44b13955235245b2497399d7a93

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Python Itertools combinations, permutations and products

```
'''
Introducing itertools
Professor Hosmer
August 2020
'''
import itertools   # Python standard library
print("Generate Combinations")
for combinations in itertools.combinations("xyz", 3):
    print(combinations)
print("Generate Permuations")
for permutations in itertools.permutations("xyz", 3):
    print(permutations)
print("Generate Product")
for variations in range(1,4):
    for products in itertools.product("xyz", repeat=variations):
        print(products)
```



```
Generate Combinations
('x', 'y', 'z')
Generate Permuations
('x', 'y', 'z')
('x', 'z', 'y')
('y', 'x', 'z')
('y', 'z', 'x')
('z', 'x', 'y')
('z', 'y', 'x')
Generate Product
('x',)
('y',)
('z',)
('x', 'x')
('x', 'y')
('x', 'z')
('y', 'x')
('y', 'y')
('y', 'z')
('z', 'x')
('z', 'y')
('z', 'z')
```

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Rainbow Table Construction for passwords

Truncated Result

…

```
'''
Simple Rainbow Table Construction
using, a dictionary
'''
import itertools   # import itertools and hashlib
import hashlib
rainbowTable = {}  # Create and empty dictionary
print("Create Simple Rainbow Table")
for variations in range(3,5):  # generate 3 and 4 char passwords
    for pwTuple in itertools.product("xyz", repeat=variations):
        pw = ""                  # initialize the pw String
        md5Hash = hashlib.md5()  # initialize the md5 object
        for eachChr in pwTuple:  # convert the tuple to string
            pw = pw+"".join(eachChr)
        pw = bytes(pw, 'ascii') # covert the string to bytes for hashing
        md5Hash.update(pw)
        md5Digest = md5Hash.hexdigest()
        rainbowTable[md5Digest] = pw
for hashValue, pwValue in rainbowTable.items():
    print(hashValue, pwValue)
```

```
Create Simple Rainbow Table
Rainbow Size:  108

f561aaf6ef0bf14d4208bb46a4ccb3ad b'xxx'
d456c1f6fa85ed2f6a26b15139c6ddc2 b'xxy'
7693cf9650737bcfc93019df022e8a0c b'xxz'
f5d8782e1d2d51da0220da48875f4186 b'xyx'
206c907786f1b442beb35b36546efd00 b'xyy'
d16fb36f0911f878998c136191af705e b'xyz'
8eab34801b8f05644302ecacb5eadc49 b'xzx'
d86ad36821fa962f47b64b2f41e9d9bb b'xzy'
8436a47abeec9845cdbd06a116cf936f b'xzz'
9cce93846bf2dc36b80183ff0378677c b'yxx'
b508680bae35614a88cafdb489b17aeb b'yxy'
9e1976ad893ad02b3159fdfe82d8f58f b'zyyz'
b9760992c468a16d01d54a29c5fad2ca b'zyzx'
d47f3c0b34a86bf7330016830a497438 b'zyzy'
33b00247e4d11b4c23a2959f4a584b2b b'zyzz'
374cf5e3c648a7c65e5c17830d3371f1 b'zzxx'
94da5e52dc74fd63ead765d4e6b9227d b'zzxy'
3b76be5134fd0a233e4c87db9c1cd944 b'zzxz'
c44a4edc249a2eb9b4b0123196d136bf b'zzyx'
ae2aeafd93fe18ebd2d1227e47a19b56 b'zzyy'
7f469dcfbd2dd43182b6f86626fc7d6c b'zzyz'
fb430ae48ecfd7436b90c312d36cc9ca b'zzzx'
02441dd7f66c49c82cef55354f467149 b'zzzy'
02c425157ecd32f259548b33402ff6d3 b'zzzz'
```

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Serialize Python Objects

```
'''
Introduction to the Pickle Library
Professor Hosmer
August 2020
'''
import pickle
print("\nSimple Pickle example\n")
# Open the destination File (write binary)
pickleFileWrite = open('examplePickle.db', 'wb')
#Create any python object
pythonList = ["abc", "def", 123, 4096, "End"]
#serialize the list and DUMP to file
print("Serializing a list:", pythonList)
pickle.dump(pythonList, pickleFileWrite)
pickleFileWrite.close()
# Open the pickle file (read binary)
pickleFileRead = open('examplePickle.db', 'rb')
# LOAD the serialized data into a list + print
print("\nLoading the pickled list\n")
retrievedList = pickle.load(pickleFileRead)
# Display the recovered List
print("Recovered List:    ", retrievedList)
pickleFileRead.close()
```

```
Simple Pickle example

Serializing a list: ['abc', 'def', 123, 4096, 'End']

Loading the pickled list

Recovered List:     ['abc', 'def', 123, 4096, 'End']
```

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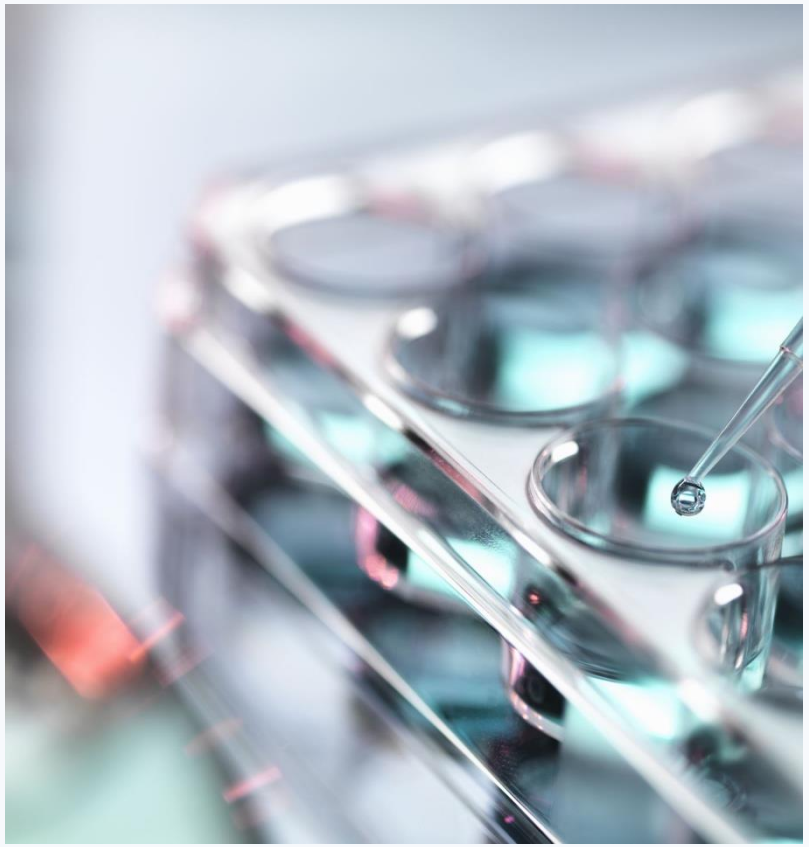
Building a mini rainbow table

- Assignment
- Using the following limited string of characters “abc123&”
- Create a rainbow table of all password 4-7 characters in length, along with the md5 hash of each password. Store them into a Python dictionary. Serialize the resulting table and store it in a file “rainbow.db”
- Display the first 5 and last 5 entries in the dictionary along with the length of the dictionary.
- You will submit 2 files.
- Your python source code
- Screenshot of your output

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Live demonstration

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Questions?

Week 5

Coming Next Week

- Creating a Python Class
- Using the Python Logging Library



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